In [1]:
```
import numpy as np
import matplotlib.pyplot as plt

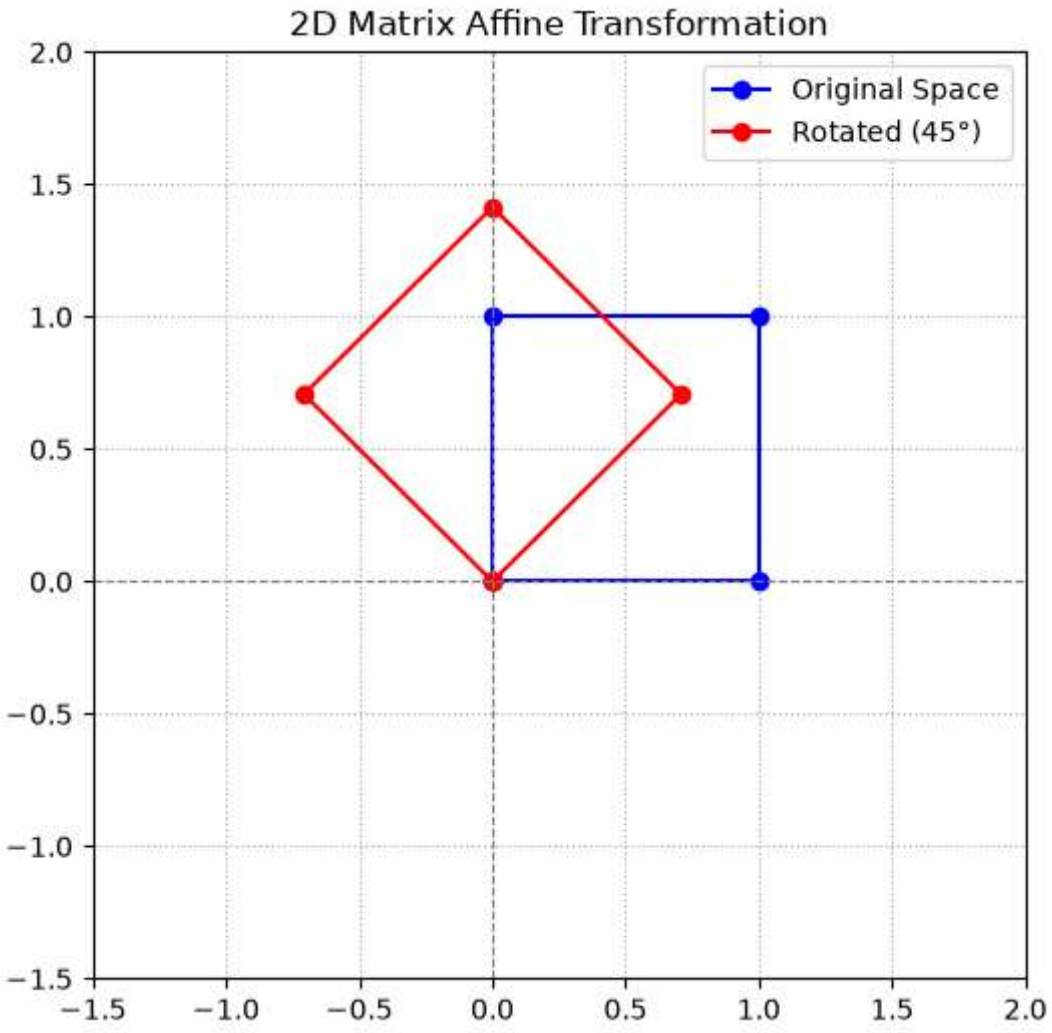
# 1. Vector Operations
v1 = np.array([2.0, 3.0])
v2 = np.array([4.0, 1.0])

dot_product = np.dot(v1, v2)
outer_product = np.outer(v1, v2)
l2_norm = np.linalg.norm(v1)

print(f"Dot Product: {dot_product}")
print(f"Outer Product:\n{outer_product}")
print(f"L2 Norm of v1: {l2_norm:.4f}\n")
W = np.array([[0.8, -0.5],
              [0.3,  0.9],
              [-0.2, 0.4]])
# Batch of 4 inputs (2 features each)
X = np.array([[1.0, 2.0],
              [3.0, 4.0],
              [-1.0, 0.5],
              [2.0, -1.5]]).T  # Shape: (2, 4)
# Bias vector (3, 1)
b = np.array([[0.1],
              [-0.2],
              [0.05]])
Z = np.dot(W, X) + b
print(f"Linear Output Shape: {Z.shape}")
print(f"Z:\n{Z}\n")
theta = np.pi / 4  # 45 degrees
A_rot = np.array([[np.cos(theta), -np.sin(theta)],
                  [np.sin(theta),  np.cos(theta)]])
pts = np.array([[0, 0], [1, 0], [1, 1], [0, 1], [0, 0]]).T
transformed_pts = A_rot @ pts
plt.figure(figsize=(6, 6))
plt.plot(pts[0, :], pts[1, :], 'b-o', label='Original Space')
plt.plot(transformed_pts[0, :], transformed_pts[1, :], 'r-o', label='Rotated (45°)'
plt.axhline(0, color='gray', linestyle='--', linewidth=0.8)
plt.axvline(0, color='gray', linestyle='--', linewidth=0.8)
plt.grid(True, linestyle=':')
plt.xlim(-1.5, 2)
plt.ylim(-1.5, 2)
plt.title("2D Matrix Affine Transformation")
plt.legend()
plt.show()
```

```
Dot Product: 11.0
Outer Product:
[[ 8.  2.]
 [12.  3.]]
L2 Norm of v1: 3.6056

Linear Output Shape: (3, 4)
Z:
[[-0.1   0.5  -0.95  2.45]
 [ 1.9   4.3  -0.05 -0.95]
 [ 0.65  1.05  0.45 -0.95]]
```




In [ ]: